

PATH-SPACE COTANGENT RIGIDITY, UNIVERSAL CONTRACTIONS, AND TANGENT REPRESENTATIONS

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ABSTRACT. Equality of invariant integrals is quotiented only by coboundaries, whereas normalized pressure equivalence also quotients constants. Weighted Folner averaging turns separating Radon functionals into invariant measures. Exact likelihood martingales and optional projections use the A4 coarse-history filtration. Source variation is formulated as a covariant derivative on the Doob Hilbert bundle and includes eigenfunction, invariant-measure, pairing, and projection derivatives. Linear and nonlinear contractions are distinguished and the full Frechet chain rule is retained.

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1. ROUND-SEVEN RECONSTRUCTION: TWO NULL SPACES, COARSE-HISTORY OPTIONAL PROJECTIONS, AND A COVARIANT DOOB CONNECTION

This section replaces the round-six controlling module. Equality of invariant integrals is separated from pressure equivalence modulo additive constants. Invariant separators are obtained by an explicit Folner averaging theorem on a weighted Radon dual. Optional projections use the coarse

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A4 filtration. When the source changes the Doob data, differentiation is covariant and includes the eigenfunction, invariant measure, inner product, and projection.

1.1. Two equivalence relations. Let $\mathcal{A}_W = C_{W,0} \cap C_{\text{dyn}}^\eta$ be the weighted dynamically Holder algebra. Define

$$\mathcal{N}_{\text{int}} = \overline{\{U - U \circ \Theta_t : U \in \mathcal{A}_W, t \geq 0\}},$$

$$\mathcal{N}_{\text{press}} = \mathbb{R}\mathbf{1} + \mathcal{N}_{\text{int}}.$$

Constants are absent from $\mathcal{N}_{\text{int}}$ and present in $\mathcal{N}_{\text{press}}$.

Theorem 1.1 (Integral and pressure cotangents). *For $F, G \in \mathcal{A}_W$ the following are equivalent:*

- (i) $\nu(F) = \nu(G)$ for every invariant probability with finite W moment;
- (ii) $F - G \in \mathcal{N}_{\text{int}}$.

Moreover F and G define the same normalized equilibrium phases and the same pressure differences if and only if $F - G \in \mathcal{N}_{\text{press}}$. If $G = F + c$, then $P(G) = P(F) + c$ and invariant integrals differ by c .

Proof. Every coboundary has zero invariant integral, proving one inclusion. For the converse suppose $F - G$ is not in $\mathcal{N}_{\text{int}}$. Hahn–Banach gives a continuous functional on $C_{W,0}$ annihilating the coboundary closure and not $F - G$. The weighted Riesz theorem represents it by a finite signed Radon measure. Lemma 1.2 below makes it invariant without changing the pairing with $F - G$. Jordan decomposition and normalization then produce invariant probabilities separating the two integrals. Adding constants gives the pressure statement and explicitly accounts for the constant shift. $\square$

1.2. Invariant averaging in the weighted Radon dual. Let μ be a finite signed measure with $\int W d|\mu| < \infty$. For the continuous-time action define the Cesaro averages

$$\mu_T = \frac{1}{T} \int_0^T (\Theta_t)_\# \mu dt.$$

Assume the platform rate–moment containment

$$\sup_{0 \leq t \leq T} \int W \circ \Theta_t d|\mu| \leq C_T \int (1 + W) d|\mu|$$

and the compact-sublevel property of W .

Lemma 1.2 (Weighted Folner averaging). *The family μ_T is weak-star relatively compact on every bounded rate–moment set. Every limit is invariant. If a functional annihilates all coboundaries, then its Cesaro averages have the same value on every $F \in \mathcal{A}_W$. Hence a separating functional has an invariant weighted Radon separator.*

Proof. The moment estimate and compact sublevels give weighted tightness. For fixed s ,

$$(\Theta_s)_\# \mu_T - \mu_T = T^{-1} \left(\int_T^{T+s} - \int_0^s \right) (\Theta_t)_\# \mu dt,$$

which tends to zero weak-star. Thus every limit is invariant. If the original functional vanishes on $F - F \circ \Theta_t$, its pairing with every translate of F equals its pairing with F , so averaging does not alter that pairing. $\square$

For Sinai platforms the rate-moment estimate follows from A3's canonical compactification; for hard spheres it follows from B2's conserved Maxwellian moment and contact entropy. These are proved inputs rather than hypotheses inserted only in C2.

1.3. Exact finite likelihoods on the coarse filtration. Let $\mathcal{F}_t^s$ be the coarse A4 history filtration and let Φ be a bounded terminal functional. Define

$$M_t^\varepsilon = \frac{\mathbb{E}[\exp\{\theta\Phi(X_T^\varepsilon)\} \mid \mathcal{F}_t^s]}{\mathbb{E} \exp\{\theta\Phi(X_T^\varepsilon)\}}.$$

Lemma 1.3 (Exact history likelihood). *The process M_t^ε is a positive mean-one martingale. Its kernel representation uses the A4 coarse-history transition kernel. If one instead conditions on the full deterministic microscopic history, the conditional expectation is the terminal functional evaluated on the unique future and no nondegenerate diffusion-kernel statement is made.*

Proof. This is the tower property of conditional expectation. The distinction of filtrations is part of the definition, not an after-the-fact reinterpretation. $\square$

Theorem 1.4 (Optional-projection convergence). *Under the A4 compact coarse-history conditional invariance principle, M^ε converges in path law and stably with respect to bounded initial coarse-history variables to*

$$M_t = \frac{\mathbb{E}[\exp\{\theta\Phi(X_T)\} \mid \mathcal{F}_t^X]}{\mathbb{E} \exp\{\theta\Phi(X_T)\}}.$$

The convergence includes the filtrations through convergence of the conditional kernels; ordinary weak path convergence alone is not used to pass conditional expectations.

Proof. Write the finite optional projection as the integral of the terminal functional against the A4 coarse-history kernel. Uniform kernel convergence on compact histories gives convergence there. Exponential containment removes the complement. Multiplication by a bounded initial-history variable is absorbed in the same kernel integral, proving stable convergence. The limiting kernel is the diffusion kernel and hence the displayed optional projection. $\square$

1.4. Fixed-Doob memory and its covariant source derivative. For a source parameter η , let h_η , λ_η , and ν_η^D be the Doob eigenfunction, eigenvalue, and invariant probability. Let $\mathcal{H}_\eta = L^2(\nu_\eta^D)$ and let P_η be the orthogonal projection onto the resolved subspace transported by multiplication with h_η . The family $\{\mathcal{H}_\eta\}$ is a Hilbert bundle. Define its connection by

$$\nabla_\eta u = \partial_\eta u + \frac{1}{2} \partial_\eta \log \frac{d\nu_\eta^D}{d\nu_0^D} u - h_\eta^{-1}(\partial_\eta h_\eta)u.$$

This connection differentiates both the inner product and the Doob conjugation. The projection derivative is the covariant derivative $\nabla_\eta P_\eta$ determined by $P_\eta^2 = P_\eta$ and orthogonality.

For fixed η define

$$C_\eta(z) = P_\eta(z - L_\eta^D)^{-1}P_\eta, \quad \widehat{K}_\eta(z) = C_\eta(z)^{-1} - zP_\eta - A_\eta.$$

The A4 Riesz–Schur dilation supplies the finite zero modes and a zero-free residual inverse on its domain.

Theorem 1.5 (Covariant Doob-memory response). *For a bounded source insertion V_η , the covariant derivative of the Doob semigroup is*

$$\nabla_\eta e^{tL_\eta^D} = \int_0^t e^{(t-s)L_\eta^D} (\nabla_\eta L_\eta^D) e^{sL_\eta^D} ds,$$

where $\nabla_\eta L_\eta^D$ contains the derivatives of the eigenvalue and eigenfunction. The derivative of the compressed resolvent and memory also contains $\nabla_\eta P_\eta$ and the derivative of the Hilbert pairing. With these terms included, pressure differentiation, Doob response, and memory compression commute as covariant maps of the Hilbert bundle.

Proof. Trivialize the Hilbert bundle by the unitary density square root and the Doob multiplication. In that fixed Hilbert space ordinary Duhamel differentiation applies. Transforming back produces the connection terms written above. Differentiate $P_\eta^2 = P_\eta$ and the orthogonality relation to obtain $\nabla_\eta P_\eta$. The resolvent identity then gives

$$\nabla C = P(z - L)^{-1}(\nabla L)(z - L)^{-1}P + (\nabla P)(z - L)^{-1}P + P(z - L)^{-1}(\nabla P).$$

Differentiating the inverse and the Riesz–Schur block formula proves the memory statement. $\square$

1.5. Girsanov and the limiting filtration. Assume the limiting diffusion has bounded C^3 coefficients and uniformly nondegenerate covariance on the resolved range. Let u solve the backward Kolmogorov equation for the terminal exponential payoff.

Theorem 1.6 (Microscopic likelihood to diffusion Girsanov). *The limit in Theorem 1.4 is*

$$M_t = \exp \left\{ \int_0^t \sigma^T \nabla \log u(s, X_s) dW_s - \frac{1}{2} \int_0^t |\sigma^T \nabla \log u|^2 ds \right\}.$$

Under the changed law, the diffusion drift receives the usual $\sigma\sigma^T\nabla\log u$ correction. The associated logarithmic value solves the risk-sensitive BSDE. This conclusion is tied to the proved coarse filtration convergence.

Proof. The classical solution is bounded away from zero by the bounded terminal payoff. Itô's formula gives the stochastic exponential. Uniform integrability comes from bounded gradient on the finite horizon. Girsanov's theorem gives the drift and the logarithmic transform gives the BSDE. $\square$

1.6. Linear and nonlinear contractions. For a continuous linear contraction C , the pulled-back source is $C^*\eta$. For a C^2 nonlinear contraction $\mathcal{C}$, use the first and second Frechet derivatives:

$$\begin{aligned} D(Q \circ \mathcal{C}^*) &= DQ DC^*, \\ D^2(Q \circ \mathcal{C}^*) &= DC D^2Q DC^* + DQ D^2\mathcal{C}^*. \end{aligned}$$

The second term is retained.

Theorem 1.7 (Typed contraction functor). *Pressure, rate, and covariance commute with continuous linear contractions on their proved source charts. They commute with nonlinear C^2 contractions through the full chain rule above whenever the pullback remains in the local analytic domain and the map is exponentially good on rate sublevels. No adjoint is assigned to a nonlinear map.*

Proof. The linear statement is the exact finite-volume identity followed by the contraction principle and source derivative convergence. The nonlinear statement follows from exponential goodness, the ordinary Frechet chain rule, and the displayed second derivative. Domain preservation is an explicit hypothesis verified for the admitted platform maps. $\square$

Theorem 1.8 (Round-seven C^2 closure). *Invariant-integral equality is quotiented only by coboundaries, while pressure equivalence also quotients constants; invariant separators are weighted Radon measures obtained by Folner averaging; finite likelihoods and optional projections use the A_4 coarse filtration; source-dependent memory response is covariant in the Doob Hilbert bundle; and nonlinear contractions use the full Frechet chain rule.*

Proof. Combine Theorems 1.1, 1.4, 1.5, 1.6, and 1.7, together with Lemmas 1.2 and 1.3. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are

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